

Ethical AI in Recommendations: A Public Good Dilemma

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Abstract

Recommender systems increasingly shape what we see and do online, effectively deciding which content is “right” for each user. Yet we still know little about what should count as the “right content” for these systems to propose, and whose values this choice should reflect. We propose a framework that treats ethical alignment in recommendations (e.g., fairness, diversity) as a public good: costly to implement, non-excludable in its benefits, and subject to free-riding. We develop a simple model in which a platform chooses between purely accuracy-driven and ethically constrained recommendations, facing users who differ in both their accuracy needs and ethical concerns. The model predicts more free-riding and weaker support for ethical alignment when individual values are *not* polarized. Building on these predictions, we design an incentivized online experiment where participants can sacrifice payoffs to influence how a recommender trades off accuracy and an ethical constraint, across treatments that mimic aligned versus polarized values. We measure willingness to pay for alignment, how it varies with polarization, its impact on outcomes, and the implications for the governance of recommender systems.

1 Introduction

Recommender systems increasingly shape what we see and do online, effectively deciding which content is “right” for each user. They determine which news stories appear in our feeds, which products and jobs we discover, and which cultural or political narratives become salient in our daily lives. While a large literature studies how to make recommendations more accurate or more engaging, much less is known about what should count as the “right content” for such systems to propose, and whose values this choice should reflect (Ge et al., 2024). In other words, beyond the technical question of how recommendations are generated lies a normative question about ethical alignment: how, and for whom, should recommender systems be aligned?

In this paper, we argue that ethical alignment in recommendations, for instance along dimensions such as fairness, diversity, or political balance, has the structure of a public good. Improving the ethical properties of a recommender typically requires costly information or effort from users: they must share personal and value-relevant data, spend time rating items or writing short reviews, and more generally provide feedback that allows the system to learn how ethical constraints should be applied.

At the same time, the benefits of alignment are largely non-excludable and non-rivalrous: once the system is more ethically aligned, many users enjoy more

trustworthy, inclusive, or balanced recommendations, including those who did not contribute. This configuration naturally generates incentives to free-ride and raises the possibility that ethical alignment will be underprovided, even when most users would prefer a more ethical system.

We study this dilemma in a setting where multiple users provide feedback to a common recommender system. Conceptually, each individual has an *ideal* composition of the content they would like to see — for example, a preferred share of politically charged versus neutral items, or of content representing specific social groups or viewpoints. At the same time, users may be willing to incur costs (in time, attention, or money) to push the system closer to their ideal, for instance by rating more items or paying for ethically constrained recommendations. From the platform’s perspective, implementing ethical constraints is also costly: fairness or diversity-based recommendation requires additional modeling, data collection and monitoring, and typically involves trade-offs with standard accuracy or engagement objectives. Real-world recommender systems operate in multi-stakeholder environments and cannot feasibly implement a fully individual-specific ethical standard. Instead, platforms choose a limited set of global or group-level constraints and thus an overall ethical level for the system, trading off user engagement against the costs of stronger constraints (such as fairness, diversity, or safeguards against harmful content).

To formalize these trade-offs, we develop a stylized theoretical model in which two user groups differ both in their ideal ethical level and in their willingness to contribute to the system. Users choose contributions that influence the platform’s choice of ethical level; the platform, observing these contributions, selects the level of alignment that maximizes its profit. In equilibrium, each user benefits from higher ethical alignment but would like others to bear the cost of inducing it. This leads to a classic public-good problem: individual contributions are below the socially efficient level, and the ethically aligned system is underprovided. The model also highlights the role of value heterogeneity and polarization: when users’ ethical ideals are close to each other, similar users may free-ride on one another, whereas when ideals are far apart, the user whose preferences are furthest from the status quo has a strong incentive to invest more in steering the system.

Building on this framework, we design an incentivized online experiment that implements a collaborative recommendation environment with a built-in public-goods structure. Participants first provide a “ground truth” ranking of movies, which defines their true preferences. They then interact with a shared recommender system over repeated rounds. In each round, they are paired with another participant and decide whether to sacrifice part of their endowment to rate a movie. Ratings are costly to the individual but improve the accuracy of the recommender for *both* members of the pair, thereby creating a social dilemma: contributing information benefits everyone, but each participant has an incentive to free-ride on their partner’s effort. We then experimentally manipulate whether participants are matched with others who share similar values or hold opposing ones. Before the interaction, we elicit participants’ positions on politically charged issues (e.g., immigration, civil rights) and on non-political, everyday preferences (e.g., dogs versus cats, reading formats). We then construct treatments that vary (i) the *domain* along which similarity is defined (political versus non-political) and (ii) the *match type* (homogeneous versus heterogeneous). In homogeneous matches, participants are paired with partners who report similar views; in heterogeneous matches, they are paired with partners whose views are dissimilar. This 2×2 design allows us to compare contribution behavior when disagreements are ideological versus neutral,

and when people cooperate with like-minded versus opposed partners.

The experimental environment maps directly into our theoretical structure. Individual contributions correspond to the effort or resources users invest in steering the system (the model’s w_i), while the accuracy of the final recommendations captures the quality of the outcome from each user’s perspective (the model’s $x_i p_i(E)$). Because each rating improves the recommender for both members of the pair, each participant’s decision to contribute generates positive externalities, and the ethical alignment or quality of the system is jointly determined. This enables us to test how value similarity, ideological polarization, and the domain of disagreement affect willingness to contribute to a shared algorithmic public good.

In line with the logic of the model, we formulate a set of pre-registered hypotheses about contribution behavior. First, (H1) we hypothesize that free-riding will be particularly pronounced when participants interact with partners who share their values: because like-minded partners have similar ethical ideals, each can expect the other to invest in shaping the system, reducing their own incentive to contribute. This leads to the prediction that, within a given domain, average contribution rates are lower in homogeneous pairs than in heterogeneous pairs. Second, (H2) we expect the effect of (dis)similarity to be stronger in the political domain than in the non-political one. Political disagreement is often more salient than disagreement over everyday preferences, so ideological heterogeneity should more strongly shape beliefs about the partner’s behavior and the perceived legitimacy of the shared system. This motivates an additional interaction hypothesis: (H3) the contribution gap between heterogeneous and homogeneous matches is larger when matching is based on political views than when it is based on non-political traits.

Beyond these primary hypotheses, the experimental design allows us to explore how contributions vary across different types of political content, how beliefs about partners’ behavior mediate contribution decisions, and how perceptions of fairness and satisfaction with outcomes depend on match type and domain. Taken together, the model and experiment provide a joint theoretical and empirical examination of ethical alignment in recommender systems as a public good: they show when and why users are willing to pay to align a system with their values, when free-riding undermines this willingness, and how value similarity and polarization shape cooperation in multi-user AI environments.

Our contribution is threefold. Conceptually, we introduce a public-good perspective on ethical alignment in recommender systems, emphasizing that alignment is not only a design choice for platforms, but also an outcome of users’ willingness to invest in shaping algorithms that serve many people at once. Theoretically, we provide a simple model of a multi-principal delegation problem in which heterogeneous users jointly incentivize an AI developer to choose an ethical standard, generating sharp comparative statics on the effects of value heterogeneity and polarization. Empirically, we implement a controlled, incentivized laboratory-in-the-field experiment that measures willingness to pay for alignment and cooperation in a shared recommender, with direct implications for the governance of AI systems that mediate access to information and opportunities. The study is structured as a registered report: we pre-specify the theoretical framework, experimental design, and main hypotheses, and then use the resulting data to assess whether users’ collective behavior is sufficient to sustain ethically aligned recommendation in the presence of free-riding and conflicting values.

2 Theoretical model: Recommender Systems as a Public good dilemma

We formalize the setting as a simple principal–agent game with two user groups and one platform. The two user groups (indexed by $i \in \{1, 2\}$) act as principals: they have their own ethical ideals and can invest in the system. The platform is the agent (a): it observes users’ contributions and chooses an overall ethical level for the recommender system. Users choose their contributions independently, taking the platform’s reaction as given. We study this non–cooperative interaction and relegate formal proofs to Appendix A.

Players and timing. The game unfolds in two stages:

1. **Stage 1 (users).** Each user group i chooses a contribution $w_i \in [0, \bar{w}]$. You can think of w_i as the total cost group i is willing to bear in terms of ratings, reviews, sharing value–relevant data, or paying for an “aligned” option.
2. **Stage 2 (platform).** The platform observes (w_1, w_2) and chooses an ethical level $E \in [0, 1]$. The variable E summarizes how strong the ethical constraints in the recommender are (e.g. how much weight is given to fairness, diversity, or other normative criteria).

We solve the model by backward induction: first the platform’s choice of E given (w_1, w_2) , then users’ contributions anticipating this response.

Ethical ideals and engagement. Each user group has its own view of what an “ethically aligned” system should look like. We capture this by an *ideal ethical level* $E_i^{\text{ideal}} \in [0, 1]$ for group i . For instance, one group might prefer a high share of politically themed or minority–representation movies, another a lower share; their E_i^{ideal} differ accordingly.

The probability that group i finds the system acceptable and engages with it is denoted by $p_i(E; E_i^{\text{ideal}})$. We assume that this probability is highest when the platform chooses the group’s ideal, and falls smoothly as E moves away. More precisely, we model p_i as a decreasing function of the distance $|E - E_i^{\text{ideal}}|$ and such that: $p_i \in [0, 1]$; $\frac{\partial p_i}{\partial E} > 0$ if $E < E_i^{\text{ideal}}$; if $E = E_i^{\text{ideal}}$, $p_i = 1$, if $|E - E_i^{\text{ideal}}|$ larger than some threshold, say K , then $p_i = 0$.

If group i is well served by the system, it derives benefit $x_i > 0$ from using it. Its net surplus, after paying contribution w_i , is then

$$U_i(E, w_i) = (x_i - w_i) p_i(E). \quad (1)$$

A higher ethical level E is attractive when it increases $p_i(E)$ by bringing the system closer to E_i^{ideal} , but larger w_i directly reduce the surplus $x_i - w_i$.

Platform objective and cost of ethics. The platform’s revenue comes from user engagement. A larger contribution w_i by group i can be interpreted as a higher per–engagement revenue from that group (because they give more data, ratings, or monetary support), multiplied by the probability $p_i(E)$ that they engage.

Let $\alpha \in (0, 1)$ be the share of group 1 in the population (so $1 - \alpha$ is the share of group 2). Given (w_1, w_2) and ethical level E , expected revenue is

$$R(E; w_1, w_2) = \alpha p_1(E) w_1 + (1 - \alpha) p_2(E) w_2. \quad (2)$$

Implementing stronger ethical constraints is costly for the platform: it requires additional modelling, data collection, monitoring, and may come with accuracy or engagement losses. We capture this with a convex cost function $C(E)$ that is increasing in E :

$$C'(E) > 0, \quad C''(E) > 0.$$

Intuitively, moving from “no ethical constraint” to “some” ethics is already costly, and pushing ethics further (closer to 1) becomes increasingly expensive.

The platform’s (agent’s) profit is therefore

$$\Pi_a(E; w_1, w_2) = \alpha p_1(E) w_1 + (1 - \alpha) p_2(E) w_2 - C(E). \quad (3)$$

Stage 2: the platform’s choice of E . Given contributions (w_1, w_2) , the platform chooses E to maximize (3). Under our assumptions on $p_i(\cdot)$ and $C(\cdot)$, this yields a unique optimal ethical level $E^*(w_1, w_2)$ characterized by the first-order condition

$$\alpha w_1 p_1'(E^*) + (1 - \alpha) w_2 p_2'(E^*) = C'(E^*). \quad (4)$$

The left-hand side is the marginal gain from raising E (higher alignment increases engagement for both groups, scaled by their contributions), and the right-hand side is the marginal cost. An increase in any contribution w_i shifts up the marginal-gain term, so the resulting best response $E^*(w_1, w_2)$ is increasing in each w_i : higher contributions by either group lead the platform to choose a higher ethical level.

Stage 1: users’ non-cooperative contributions. At Stage 1, each user group chooses w_i anticipating the platform’s reaction $E^*(w_1, w_2)$. Substituting $E^*(w_1, w_2)$ from (4) into (1), user i solves

$$\max_{w_i \in [0, \bar{w}]} U_i(E^*(w_i, w_j), w_i) = (x_i - w_i) p_i(E^*(w_i, w_j)),$$

taking the other group’s contribution w_j as given.

Differentiating with respect to w_i , an interior best response satisfies

$$-p_i(w_i^*) + (x_i - w_i) p_i'(E^*) \frac{\partial E^*}{\partial w_i} = 0. \quad (5)$$

The first term is the direct marginal cost of contributing a bit more (it reduces the surplus $x_i - w_i$), while the second term captures the marginal benefit: increasing w_i pushes the platform to choose a higher E^* (since $\partial E^* / \partial w_i > 0$), which in turn raises $p_i(E^*)$.

A non-cooperative equilibrium is a pair (w_1^*, w_2^*) such that each w_i^* solves this problem, given w_j^* . Under mild regularity conditions (strict concavity of the objective in w_i), we show in Appendix A that such an equilibrium exists and is unique.

In equilibrium, both types of principal are affecting the **same ethical choice**. Their incentive need to balance, i.e.,

$$\frac{\partial E}{\partial w_1} = \frac{\partial E}{\partial w_2}$$

Hence

$$\frac{(x_2 - w_2)}{(x_1 - w_1)} = \frac{\epsilon_{p_1}}{\epsilon_{p_2}}$$

This equilibrium condition has a clear economic interpretation: the relative surplus each user is willing to forgo (i.e., $x_i - w_i$) must balance their respective

elasticities of engagement with respect to the ethical level E . That is, users who are more sensitive to ethical misalignment, represented by a higher elasticity ε_{p_i} of click probability, must offer higher incentives to induce the developer to choose a more aligned ethical standard. Intuitively, the system reaches stability only when the marginal willingness to pay of each user reflects how quickly they would disengage if ethical quality deteriorates. This captures the idea that ethical behavior by the developer is a public good: users benefit jointly from a more ethical system, but must contribute individually to incentivize it, leading to potential free-riding and underprovision in equilibrium.

The key feature of this equilibrium is that each group internalizes only *its own* benefit from raising E^* : when choosing w_i , group i ignores the fact that a higher E^* also benefits the other group j . Because each contribution w_i pushes E^* up for everyone, contributions have the structure of a public good, and in equilibrium they are lower than what would be chosen by a benevolent planner who maximizes $U_1 + U_2$. This underprovision of contributions translates into underprovision of ethical alignment in the recommender system: the platform chooses an ethical level E^* that is too low from a collective point of view.

3 Experimental design

Our theoretical framework is two-sided (users and developer). In the experiment, we hold the developer-side learning rule fixed *by design* to isolate the *user-side* mechanism: how ethical alignment—especially under polarization—shapes willingness to take privately costly actions that improve a shared recommendation system, even when the benefits are partially non-excludable and free-riding is possible. Specifically, we ask:

- Does perceived ethical alignment increase willingness to invest in shared algorithmic quality?
- Are participants more likely to free-ride when paired with ideologically misaligned partners?
- Does perceived ethical similarity enhance trust and coordination in digital environments?

3.1 Experimental Procedure

The experimental procedure has three phases: (i) a pre-treatment survey and preference elicitation to obtain a participant-specific “ground truth”; (ii) repeated rating rounds with stranger matching to study costly individual contributions to a shared recommender system; and (iii) a prediction and payment stage that links payoffs to recommendation quality and thereby incentivizes truthful and attentive participation (Fig. 1). Treatments vary along two dimensions: the *domain of disagreement* used to define similarity (political vs. non-political) and the *match type* (homogeneous vs. heterogeneous) (Table 1).

Participants start with a fixed endowment that can be spent during the task. In each round, they decide whether to incur a small private cost to rate a movie. Submitting a rating provides information to the system and improves its ability to predict the participant’s own preferences in the final stage. Crucially, learning is *shared*: additional data improves the system for both members of the pair. This creates a standard public-goods problem in a recommender setting: contributing is privately costly, but it generates a partially non-excludable benefit, so participants

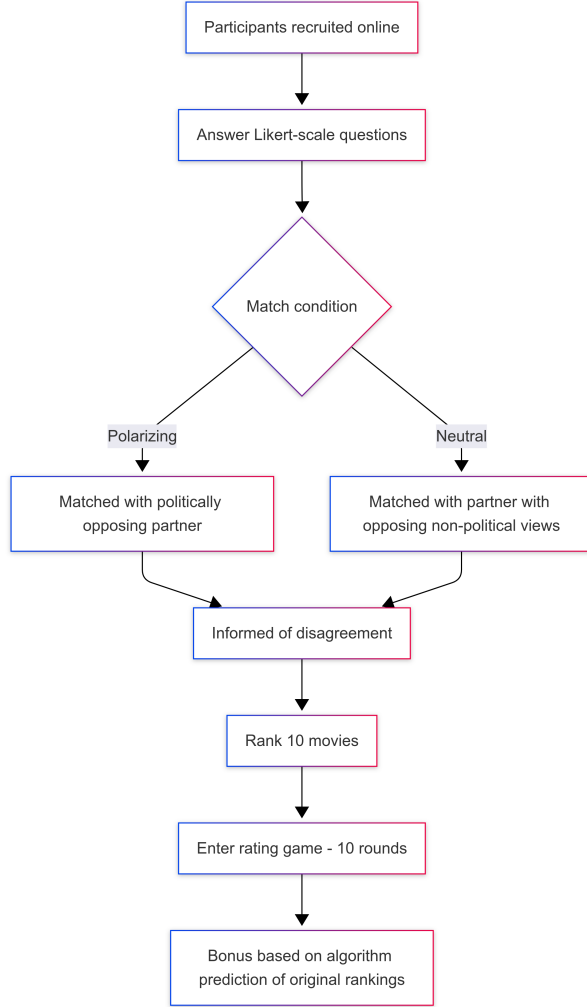


Figure 1: Caption

can be tempted to free-ride on their partner’s effort. Full instructions are provided at the end of the paper.

To implement this in a fully transparent way, without programming the developer-side, and avoiding deception, we use a simple, pre-specified fixed learning rule that maps *total* contributions (see below) into recommendation accuracy in the final prediction stage. Because the contribution-to-accuracy mapping is identical in all conditions, any treatment differences in contribution behavior reflect user-side responses to the matching context—political (polarized) vs. non-political (neutral), and homogeneous vs. heterogeneous—rather than differential algorithmic behavior. This matching is our empirical way to vary ethical alignment across users. This means treatment effects are conservative: they are a lower bound given fixed technical conditions.

Phase 1 — Pre-Treatment Survey & Preference Elicitation. Participants are recruited via Prolific. Before the task, they complete a short Likert-scale survey covering *political issues* (e.g., abortion, immigration, gun control) and *non-political preferences* (e.g., eBooks vs. print; dogs vs. cats).¹ Participants are in-

¹Political orientation measures follow Aybar et al. (2024), Yeung and Quek (2025), and the ANES 1–7 ideology scale. References for the privacy questionnaire to be added.

formed that truthful answers matter because they determine subsequent matching and monetary payoffs. In our theoretical framework, these survey responses proxy participants’ ethical ideals.

Using these responses, we thus define similarity and dissimilarity for matching in each experimental condition. Participants then provide a ground-truth ranking by ordering a curated list of $K = 5$ movies from the most to the least preferred. This ground-truth ranking is used later to score recommendation accuracy and determine the final bonuses (see Phase 3).

Phase 2 — Rating Rounds (Public-Goods Contribution). Each participant receives an endowment of $e = \text{€}2.50$. Over $R = 10$ rounds, a stranger-matching protocol pairs participants with different partners. In each round, participants decide simultaneously whether to:

- pay $c = \text{€}0.25$ to submit a rating (i.e., contribute training data), or
- skip at zero cost.

Each rating (by either player) increases the recommender’s learning parameter by $a = 0.10$, so that total learning depends on *both* players’ contributions. This generates a public-goods dilemma: contributing is privately costly but partially non-excludable because the partner benefits from the improved recommender even if they do not contribute.

To make the social context salient, participants are told whether the partner they are matched with in that round is *similar* or *opposed* based on the pre-treatment survey depending on the treatment. We only provide this coarse similarity/opposition label. Participants never see the partner’s identity. In addition, we avoid describing outcomes in terms of probabilities. Instead, we explain both accuracy and payoffs in concrete terms as the recommender system’s ability to *guess* the participant’s ground-truth ranking: how many positions in the 5-movie ranking the system gets exactly right. More total ratings (from the participant and the partner) increase the system’s ability to guess these positions correctly, which increases expected bonuses (see Phase 3). Importantly, movie content is held constant across treatments: in every condition, participants are shown an equal mix of political and non-political movies (50/50), drawn from the same pool and randomized across rounds. See Appendix for the movie pool classification.

Phase 3 — Prediction & Payment (Poisson- K scoring). At the end of the 10 rounds, the system produces a predicted ranking for each participant. Instead of deterministically computing accuracy from a black-box algorithm, we use a transparent stochastic scoring rule based on the number of correctly guessed positions K .² At baseline (before any ratings are provided), the system is calibrated to get about 1 out of 5 ranking positions right on average. Under full mutual rating, the learning parameter increases and the system typically gets about 3 out of 5 positions right on average).³

²We draw K from a truncated Poisson

³Learning rule. Let $r_i \in \{0, \dots, 10\}$ be the number of ratings submitted by participant i and r_j the number submitted by their partner (over the same 10 rounds). The mechanism sets

$$\lambda = 1 + a(r_i + r_j), \quad a = 0.10.$$

Thus, baseline accuracy corresponds to $\lambda = 1$ (roughly ≈ 1 correct position out of 5 on average), and under full mutual rating $r_i = r_j = 10$, $\lambda = 3$ (roughly ≈ 3 correct positions out of 5 on average; truncated mean $E[K] \approx 2.67$).

Participants earn a bonus that is linear in realized accuracy: €2 for each correctly guessed position, where $K \in \{0, 1, 2, 3, 4, 5\}$ is the number of correct positions:

$$\text{Bonus}_i = b K_i, \quad b = \text{€}2,$$

so the maximum bonus is €10 when $K_i = 5$. Final payoff is:

$$\Pi_i = e - c r_i + b K_i, \quad e = \text{€}2.50, \quad c = \text{€}0.25, \quad b = \text{€}2.$$

This payoff rule guarantees no losses.⁴

3.2 Experimental Treatments

The design is a 2×2 factorial that varies: (i) *domain of disagreement* used to define similarity (political vs. non-political), and (ii) *match type* (homogeneous = similar vs. heterogeneous = dissimilar). Assignments are implemented at the round level via stranger matching and the survey-based similarity metric.

In political treatments, participants are matched based on similarity or dissimilarity in their political survey responses. In non-political treatments, pairing depends on everyday, apolitical preferences. This design allows us to compare behavior when disagreement is ideological versus neutral, and when partners share versus oppose each other’s views (see Table 1).

Table 1: Experimental Conditions

Domain of Disagreement	Match Type	
	Homogeneous	Heterogeneous
Non-political	Neutral–Homogeneous	Neutral–Heterogeneous
Political	Political–Homogeneous	Political–Heterogeneous

3.3 Hypotheses and analysis plan

In the theoretical model, users benefit when the recommender delivers outputs that are both (i) accurate with respect to their preferences and (ii) ethically aligned with their ideals. This is summarized on the developer side by a system choice E (the ethical/quality level of the recommender), which increases the probability that a user receives a recommendation that matches their ideal and therefore yields utility.

In the experiment, we capture this channel in reduced form. Costly user contributions increase the system’s learning parameter (λ), which improves prediction quality in the final stage, measured by the realized number of correctly guessed positions (K) in the 5-movie ranking; higher K implies higher payoffs. Learning is partially non-excludable because λ depends on ratings provided by both members of the pair, so each participant can benefit from the other’s contribution, creating incentives to free-ride. Finally, our matching treatments operationalize variation in ethical alignment between users: homogeneous matches pair more aligned participants, whereas heterogeneous matches pair less aligned participants (in the political domain, this corresponds to polarized pairings).

⁴the minimum possible payoff occurs when a participant rates in all 10 rounds and realizes $K = 0$,

$$\Pi_{\min} = e - 10c = 2.50 - 10 \times 0.25 = 0.$$

A single rating increases each player’s expected bonus by approximately ba but costs the rater c . A strict Prisoner’s Dilemma requires

$$ba < c < 2ba.$$

Intuitively, this means that the private return from rating (the increase in the rater’s expected bonus, ba) is smaller than the private cost c , so rating is not individually optimal; however, the total return to the pair (the increase in both players’ expected bonuses, $2ba$) exceeds the cost, so mutual rating is socially efficient. With $b = \text{€}2$ and $a = 0.10$, we have $ba = \text{€}0.20$ and $2ba = \text{€}0.40$, and since $c = \text{€}0.25$, defection is individually dominant, yet mutual contribution increases both accuracy and total payoffs.

3.3.1 Experimental Hypotheses

Our primary behavioral outcome is contribution, measured as the number of rounds in which a participant chooses to rate. Let $c_{ir} \in \{0, 1\}$ equal 1 if participant i rates in round $r \in \{1, \dots, 10\}$. Define

$$C_i = \sum_{r=1}^{10} c_{ir} \in \{0, 1, \dots, 10\}.$$

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Let $t \in \{\text{NP-Het}, \text{NP-Hom}, \text{P-Het}, \text{P-Hom}\}$ index the four conditions (non-political vs. political domain, crossed with heterogeneous vs. homogeneous matching). Let $\mathcal{I}_t$ be the set of participants in treatment t , and define the mean contribution count

$$\bar{c}_t = \frac{1}{|\mathcal{I}_t|} \sum_{i \in \mathcal{I}_t} C_i.$$

H1 – Heterogeneous matches increase contribution (within each domain). We expect participants in heterogeneous matches to contribute more (rate in more rounds) than participants in homogeneous matches, both in the non-political and in the political domain.

$$\begin{aligned} \text{Non-political domain:} & \quad \begin{cases} H_0^{\text{NP}} : \bar{c}_{\text{NP-Het}} - \bar{c}_{\text{NP-Hom}} \leq 0 \\ H_1^{\text{NP}} : \bar{c}_{\text{NP-Het}} - \bar{c}_{\text{NP-Hom}} > 0 \end{cases} \\ \text{Political domain:} & \quad \begin{cases} H_0^{\text{P}} : \bar{c}_{\text{P-Het}} - \bar{c}_{\text{P-Hom}} \leq 0 \\ H_1^{\text{P}} : \bar{c}_{\text{P-Het}} - \bar{c}_{\text{P-Hom}} > 0 \end{cases} \end{aligned}$$

This prediction is grounded in our theoretical model, where each user chooses a costly contribution w_i that affects the system’s quality through an aggregate channel (e.g., $E = E(w_1 + w_2)$) and the resulting improvement benefits both users. Because benefits are shared, contributions are strategic substitutes: the marginal return to w_i decreases in the expected contribution of the partner. In the experiment, we hold the learning rule fixed across treatments, so the mechanism itself

⁵As a secondary measure of low contribution/free-riding, we also define

$$FR_i = \mathbb{1}\{C_i < 5\},$$

a binary indicator equal to 1 if participant i rates in fewer than half of rounds. If preferred, depending on the pilot, we can preregister an alternative threshold (e.g., $C_i \leq 2$) and treat it as a robustness check.

does not change; what changes is the perceived alignment within the pair. As a result, $H1$ is a prediction about user responses to the matching context (perceived (mis)alignment) rather than differential algorithmic behavior. When ethical ideals are highly aligned (homogeneous pairs), participants may expect the partner to invest and therefore face stronger incentives to free-ride. When ideals are opposed (heterogeneous pairs), alignment is lower and the perceived stakes of steering the system are more asymmetric, increasing the incentive for at least one participant to contribute (see Section XX). A competing channel is that similarity may increase trust and coordination and thus increase contribution, reversing the direction predicted in $H1$.

Estimation & decision rule. For each domain separately, estimate the difference in means $\bar{c}_{\text{Het}} - \bar{c}_{\text{Hom}}$ and report a one-sided p -value from (i) a difference-in-means test or (ii) an OLS regression with a heterogeneous indicator (equivalent). Reject H_0^{NP} and/or H_0^{P} at level α if $p < \alpha$.

Interpretation (both sides).

- *Reject H_0^{NP} :* In the non-political domain, heterogeneous matches increase contribution relative to homogeneous matches.
- *Fail to reject H_0^{NP} :* No evidence that heterogeneity increases contribution in the non-political domain.
- *Reject H_0^{P} :* In the political domain, heterogeneous matches increase contribution relative to homogeneous matches.
- *Fail to reject H_0^{P} :* No evidence that heterogeneity increases contribution in the political domain.

H2 – Interaction: misalignment matters more in the political (polarized) domain. The heterogeneous–homogeneous contribution gap is larger in the political domain than in the non-political domain. The heterogeneous–homogeneous contribution gap is larger in the political domain than in the non-political domain. Since the algorithm is identical in all conditions, this is a framing effect, and any estimated interaction is a lower bound

Definition.

$$\Delta_{\text{NP}} = \bar{c}_{\text{NP-Het}} - \bar{c}_{\text{NP-Hom}}, \quad \Delta_{\text{P}} = \bar{c}_{\text{P-Het}} - \bar{c}_{\text{P-Hom}}.$$

$$\begin{cases} H_0^{\text{H2}} : \Delta_{\text{P}} - \Delta_{\text{NP}} \leq 0 \\ H_1^{\text{H2}} : \Delta_{\text{P}} - \Delta_{\text{NP}} > 0 \end{cases}$$

Estimation & decision rule. Estimate a difference-in-differences, equivalently via

$$C_i = \alpha + \beta \text{Het}_i + \delta \text{Pol}_i + \theta (\text{Het}_i \times \text{Pol}_i) + \varepsilon_i,$$

where Het_i indicates heterogeneous matching and Pol_i indicates the political domain. Test $H_0^{\text{H2}} : \theta \leq 0$ against $H_1^{\text{H2}} : \theta > 0$ using a one-sided test at level α .

Interpretation.

- *Reject H_0^{H2} :* The heterogeneity effect is larger in the political domain than in the non-political domain.
- *Fail to reject H_0^{H2} :* No evidence that the heterogeneity effect is stronger in the political domain.

It is important to note that the interpretation of $H2$ depends on the fact that the learning rule is fixed and identical across treatments. A failure to reject H_0^{H2} therefore means that we find no detectable *additional* framing (polarization) effect beyond the baseline heterogeneous–homogeneous difference. Since the recommender does not respond differently across conditions, this is a conservative (lower-bound) test of polarization under fixed technical conditions.

H3 – Contributions differ for political vs. non-political movies. Across rounds, contribution differs by movie type. We classify each title as either political ($g = P$) or non-political ($g = NP$). Because all treatments use the same movie pool and the same 50/50 split between political and non-political titles, any difference we detect here reflects how participants respond to the *content type* itself, not differences in the set of movies shown across treatments.

Definition. Let $g \in \{P, NP\}$ denote the movie type shown in a round. Define

$$C_{ig} = \sum_{r:g(r)=g} c_{ir}, \quad g \in \{P, NP\},$$

and let $\bar{c}_g$ be the corresponding sample mean.

We hypothesize

$$\begin{cases} H_0^{H3} : \bar{c}_P = \bar{c}_{NP} \\ H_1^{H3} : \bar{c}_P \neq \bar{c}_{NP} \end{cases}$$

Estimation & decision rule. Estimate a round-level regression with participant fixed effects,

$$c_{ir} = \alpha_i + \gamma \mathbb{1}\{g(r) = P\} + u_{ir},$$

with standard errors clustered at the participant level. Reject H_0^{H3} at level α if the (two-sided) p -value for γ is below α .

Pre-specified heterogeneity check (content $\times$ match type). To assess whether the content effect differs by match type, estimate

$$c_{ir} = \alpha_i + \gamma \mathbb{1}\{g(r) = P\} + \eta \left(\mathbb{1}\{g(r) = P\} \times \text{Het}_i \right) + u_{ir},$$

where Het_i equals 1 under heterogeneous matching and 0 under homogeneous matching. Here η captures whether the political–non-political contribution gap differs between heterogeneous and homogeneous matches; we will report $\hat{\eta}$ and its two-sided p -value.

H3 isolates a *content* channel: whether participants are more willing to incur the rating cost when the item itself is political rather than non-political. Because item exposure is balanced (50/50) and drawn from the same pool in every treatment, this comparison is not driven by differences in the set of movies shown across conditions. The heterogeneity check assesses whether this content effect is itself sensitive to the social context created by matching (heterogeneous vs. homogeneous), i.e., whether political content amplifies or attenuates the contribution gap across match types.

3.4 Sample size and power

We target $N = 400$ completed participants recruited on Prolific, equally allocated across the four cells of the 2×2 design ($n = 100$ per condition). The primary outcome C_i ranges from 0 to 10 and is analyzed using difference-in-means (equivalently OLS with treatment indicators).

Power calculations are reported in terms of minimum detectable effects (MDEs) under standard normal-approximation assumptions. For a two-sample comparison with n participants per group, the MDE in standardized units (Cohen’s d) for a one-sided test is

$$d_{\text{MDE}} \approx (z_{1-\alpha} + z_{1-\beta}) \sqrt{\frac{2}{n}},$$

where α is the significance level and $1 - \beta$ is power. With $n = 100$, $\alpha = 0.05$ (one-sided, so $z_{1-\alpha} = 1.645$) and 80% power ($z_{1-\beta} = 0.842$), this yields

$$d_{\text{MDE}} \approx (1.645 + 0.842) \sqrt{\frac{2}{100}} \approx 0.35.$$

Interpreting this in units of C_i requires an assumption on $\text{sd}(C_i)$; using a conservative benchmark $\text{sd}(C_i) \approx 3$, this corresponds to an MDE of about $0.35 \times 3 \approx 1.05$ additional rated rounds.

For the interaction in $H2$, the difference-in-differences estimand compares four cell means. With balanced cells ($n = 100$ each) and common standard deviation σ , the standard error of the interaction is approximately

$$\text{se}(\widehat{\Delta_P - \Delta_{\text{NP}}}) \approx \sqrt{\frac{4\sigma^2}{n}} = \frac{2\sigma}{\sqrt{n}} \approx 0.2\sigma \quad \text{for } n = 100.$$

Under a one-sided $\alpha = 0.05$ test with 80% power, the corresponding MDE is approximately

$$\text{MDE} \approx (z_{1-\alpha} + z_{1-\beta}) \frac{2\sigma}{\sqrt{n}} = (1.645 + 0.842) \times 0.2\sigma \approx 0.50\sigma,$$

i.e., about 1.5 rated rounds if $\sigma \approx 3$. These benchmarks will be updated once pilot data provide an empirical estimate of $\text{sd}(C_i)$ and completion/attrition rates.

Additional preregistered analysis details

Unit of analysis and standard errors. The main tests use participant-level outcomes (C_i , FR_i). If we estimate round-level specifications (e.g., c_{ir} as an outcome), we will cluster standard errors at the participant level.

Multiple hypotheses. We treat H1 and H2 as confirmatory. H3 is exploratory unless categories and comparisons are fully specified ex ante; if treated as confirmatory, we will preregister categories and apply a stated multiple-testing correction.

Reporting. For each hypothesis, we will report treatment means, estimated differences (or $\hat{\theta}$), confidence intervals, and the pre-registered one-sided (H1, H2) or two-sided (H3) p -values, together with the interpretation of rejection vs. non-rejection as stated above.

3.5 Model Extension

Ethical alignment in our setting cannot be reduced to “letting each user choose,” because the platform ultimately implements a *shared* recommendation policy (or a small set of global constraints) that jointly affects many users. Even if users have well-defined views about what counts as the “right” content, their efforts (ratings, feedback, payments, value-relevant data) primarily shift a common ethical standard rather than a purely individual one. This is where Sen’s insight is useful: once preferences are even mildly other-regarding—as ethical concerns typically are, since they involve fairness, diversity, or harm to others—combining individual decisiveness over outcomes with efficiency-style requirements (e.g., Pareto improvements) can generate conflicts in collective choice. In our framework, the implication is that ethical alignment is better modeled as a public good than as a standard preference-aggregation problem: contributions are privately costly but partially non-excludable in their benefits, which creates free-riding incentives and leads to underprovision of alignment.

A simple extension of our “preferences over ethical AI” formulation makes the strength of ethical concern explicit. Let $E \in [0, 1]$ denote the system’s ethical level and E_i^{ideal} user i ’s ideal. We can augment utility with an “ethical disutility” term weighted by an individual salience parameter $\theta_i \geq 0$:

$$U_i(E, w_i) = (x_i - w_i) p_i(E) - \theta_i d(E, E_i^{\text{ideal}}), \quad (6)$$

where $d(\cdot, \cdot)$ is a distance function (e.g., $d(a, b) = |a - b|$ or $d(a, b) = (a - b)^2$). The parameter θ_i captures how much the user values ethical alignment *per se* (beyond accuracy/engagement). This extension yields immediate comparative statics: higher θ_i increases the marginal benefit of shifting E toward E_i^{ideal} , raising willingness to contribute w_i in equilibrium, while polarization can matter both through dispersion in E_i^{ideal} and through heterogeneity in θ_i .

Example of other-regarding preferences

$$U_i(E, w_i) = (x_i - w_i) p_i(E) - \theta_i d(E, E_j^{\text{ideal}}), \quad (7)$$

My utility increases as E increases and approaches E_i^{ideal} , because it induces an increase of $p_i(E)$; but now my utility increases also as the new term $d(E, E_j^{\text{ideal}})$ decreases, that is the more E approaches E_j^{ideal} , that is we assume $\frac{\partial d}{\partial E} < 0$ as long as $E < E_j^{\text{ideal}}$. The first order condition in the first stage becomes:

$$-p_i(w_i^*) + (x_i - w_i) p_i'(E^*) \frac{\partial E^*}{\partial w_i} - \theta_i * \frac{\partial d}{\partial E} * \frac{\partial E^*}{\partial w_i} = 0.$$

The first two terms are the same as before. The third term is positive, it represents the further marginal benefit from increasing w_i that in turn implies a higher E that reduces the disutility from the distance with E_j^{ideal} . It implies that the optimal E is higher with other regarding preferences than with standard preferences.

A further natural generalization is multi-dimensional ethics. Let $E \in [0, 1]^K$ index K ethical dimensions (e.g., fairness, diversity, safety), with ideals E_{ik}^{ideal} and weights θ_{ik} :

$$U_i(E, w_i) = (x_i - w_i) p_i(E) - \sum_{k=1}^K \theta_{ik} d(E_k, E_{ik}^{\text{ideal}}). \quad (8)$$

This formulation directly connects to governance questions about which ethical dimensions are implemented globally and whose weights are reflected in the deployed constraint vector.